Lab Experiment 5

Global engineering collaboration problem (part 2):

Learning Outcomes

By the end of these activities, students will be able to:

1. Identify the differences between SI, BG, and USCS units commonly used in engineering practice.
2. Interpret mechanical drawings with missing or mixed unit systems.
3. Apply accurate unit conversions to complete engineering calculations and drawings.
4. Demonstrate problem-solving skills in real-world scenarios involving international collaboration.
5. Reflect on the importance of standardization and clear communication in global engineering projects.

Lab Exercise

Objective:

To simulate a real-world engineering scenario where students collaborate with an overseas company that uses a different system of units. Perform accurate conversions between SI and BG/USCS systems and complete the documentation to ensure clarity and accuracy.

Instructions:

Complete the following activities referring to chapter 5, the lab guide, and other appropriate resources in your lab report paper.

Lab activity overview:

The U.S. logistics company in which you work for continues working with E-MIDER in Germany. They request help evaluating the two mechanical parts your team worked on previously (lab 4).

The parts were:

- **Safety Seat Mounting Bracket**
- **Connecting Rod**

Each part has three new material variants with different costs, advantages, and drawbacks.

Safety Seat Mounting Bracket:

Material	Cost (per unit)	Pros	Cons
Mild Steel	$22.00	Strong, cheap, easy to install	Heavy, prone to corrosion
Aluminum Alloy	$38.00	Lightweight, corrosion-resistant	Higher cost, lower fatigue strength
Composite Polymer	$42.00	Very light, corrosion-proof	Expensive, harder to manufacture

Connecting Rod:

Material	Cost (per unit)	Pros	Cons
Cast Iron	$28.00	Inexpensive, durable	Heavy, brittle
Aluminum Alloy	$45.00	Lightweight, good for speed	Higher cost, wears faster
Forged Steel	$52.00	Strong, long lifespan	Expensive, harder to manufacture

In addition to the variant tables, the client also provided a list of objectives for both parts

- **Weather Resistance:** Parts must withstand outdoor exposure, rain, and humidity without major corrosion.
- **Stable Material Prices:** Material costs should not fluctuate heavily with market swings.

- **Weight Reduction:** Lighter parts reduce vehicle fuel/energy consumption.
- **Durability & Safety:** Parts must last for years in real operating conditions without critical failures.
- **Cost Efficiency:** The final product must stay affordable for U.S. customers.
- **Ease of Manufacturing:** Materials that simplify production and reduce defects are preferred.
- **Sustainability:** Preference for materials with lower environmental impact or recyclability.

Your job is to:

1. **Analyze** the options for both parts.
2. **Compare** each option against these 7 objectives in two tables (one for each part). You might want to do some research on each material in order to come up with a good comparison.
3. **Decide** which variant best balances the objectives (There are no right or wrong answers).
4. **Justify** your decision in a **formal business letter created in Microsoft Word**, using the formatting and documentation tools provided in the guide and chapters for this module.

The requirements for the formal business letter are as follows:

Document Setup:

- Final document must have the following header and footer:
  Header: Course name and number (TECH 100 – Intro to engineering & Technology)
  Footer: Left = "By [Student Name]", **Center** = "lab #5 – Business letter",

  Right = page number
- Insert current date at the top (Use insert > Date & Time)

Formatting & Style:

- Use **Times New Roman** or **Calibri**, 12pt font, single-spaced.
- Apply Styles for:
  - Company name & Address = Title style

    - Greeting line = **Strong** style
    - Closing line = *Emphasis* style
- Body text must be justified (not left aligned).
- Add *Intense Quote style* for one key sentence (e.g., "*We recommend forged steel for its superior durability*")

Content Structure:

- Your letter must follow this structure:
    - Opening: Formal greeting
    - Introduction: State the purpose of the letter.
    - Main Body:
        1. Insert both **comparison tables**,
        2. Insert a **bulleted list** (diamond bullets) of the pros and cons for your chosen option for each part.
    - Conclusion: Recommendation and justification.
    - Closing: Formal signature line.

Reflection and transparency (for your lab report)

Prepare a brief reflection that includes:

- The AI tool used and how (If you used one)
- What you learned while doing this lab

Grading Rubric

- **Objectives:** Identifies objectives and provides the necessary background or context (0-3 points)
- **Results/Analysis:** Presents results clearly. Analyzes findings correctly, with proper calculations and/or comparisons as needed. (0-6 points)
- **Discussion & Reflection:** Explains significance of results, discusses errors and connects outcomes to engineering/technology concepts. (0-5 points)
- **Professionalism & Format:** Report is well-organized, follows the provided template, uses clear language, proper grammar and formatting. Includes citation and references (if outside sources were used) (0-6 points)
- **Total:** 20 points

Deliverables

You must submit a lab report following the lab template. The lab report must include the work done for the first three job questions as well as the usual reflection at the end, plus applicable references.

Your job is to:

1. **Analyze** the options for both parts.
2. **Compare** each option against these 7 objectives in two tables (one for each part). You might want to do some research on each material in order to come up with a good comparison.
3. **Decide** which variant best balances the objectives (There are no right or wrong answer).

In addition to the report, you must also submit your business letter in both .**pdf** and .docx **format**

